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978-0-521-85757-4 - Geometric Folding Algorithms: Linkages, Origami, Polyhedra

Erik D. Demaine and Joseph O'Rourke

Frontmatter

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GEOMETRIC FOLDING ALGORITHMS

Folding and unfolding problems have been implicit since Albrecht Dürer in the early 1500s but have only recently been studied in the mathematical literature. Over the past decade, there has been a surge of interest in these problems, with applications ranging from robotics to protein folding. With an emphasis on algorithmic or computational aspects, this comprehensive treatment of the geometry of folding and unfolding presents hundreds of results and more than 60 unsolved “open problems” to spur further research.

The authors cover one-dimensional (1D) objects (linkages), 2D objects (paper), and 3D objects (polyhedra). Among the results in Part I is that there is a planar linkage that can trace out any algebraic curve, even “sign your name.” Part II features the “fold-and-cut” algorithm, establishing that any straight-line drawing on paper can be folded so that the complete drawing can be cut out with one straight scissors cut. In Part III, readers will see that the “Latin cross” unfolding of a cube can be refolded to 23 different convex polyhedra.

Aimed primarily at advanced undergraduate and graduate students in mathematics or computer science, this lavishly illustrated book will fascinate a broad audience, from high school students to researchers.

Erik D. Demaine is the Esther and Harold E. Edgerton Professor of Electrical Engineering and Computer Science at the Massachusetts Institute of Technology, where he joined the faculty in 2001. He is the recipient of several awards, including a MacArthur Fellowship, a Sloan Fellowship, the Harold E. Edgerton Faculty Achievement Award, the Ruth and Joel Spira Award for Distinguished Teaching, and the NSERC Doctoral Prize. He has published more than 150 papers with more than 150 collaborators and coedited the book *Tribute to a Mathemagician* in honor of the influential recreational mathematician Martin Gardner.

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Geometric Folding Algorithms

Linkages, Origami, Polyhedra

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To my father, Martin Demaine
– Erik

To my mother, Eleanor O'Rourke
– Joe

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Preface

At how many points must a tangled chain in space be cut to ensure that it can be completely unraveled? No one knows. Can every paper polyhedron be squashed flat without tearing the paper? No one knows. How can an unfolded, precreased rectangular map be refolded, respecting the creases, to its original flat state? Can a single piece of paper fold to two different Platonic solids, say to a cube and to a tetrahedron, without overlapping paper? Can every convex polyhedron be cut along edges and unfolded flat in one piece without overlap? No one knows the answer to any of these questions.

These are just five of the many unsolved problems in the area of geometric folding and unfolding, the topic of this book. These problems have the unusual characteristic of being easily comprehended but they are nevertheless deep. Many also have applications to other areas of science and engineering. For example, the first question above (chain cutting) is related to computing the folded state of a protein from its amino acid sequence, the venerable “protein folding problem.” The second question (flattening) is relevant to the design of automobile airbags. A solution to the last question above (unfolding without overlap) would assist in manufacturing a three-dimensional (3D) part by cutting a metal sheet and folding it with a bending machine.

Our focus in this book is on geometric folding as it sits at the juncture between computer science and mathematics. The mathematics is mainly geometry and discrete mathematics; the computer science is mainly algorithms, more specifically, computational geometry. The objects we consider folding are 1D linkages, 2D paper, and the 2D surfaces of polyhedra in 3-space, and thus our title: *Geometric Folding Algorithms: Linkages, Origami, Polyhedra*. As we devote Chapter 0 to a foretaste of the topics covered, we restrict ourselves here to meta-issues.

Because the topics are tangible, physical intuition makes them accessible to those with a wide variety of preparations. Indeed, we have had success using this material to teach mathematical and computational concepts at the high-school, middle-school, and even grade-school levels. However, our presentation in this book is pitched more at the college/graduate-school levels, and we believe professional researchers will find much of interest as well. Because the topics are so variegated, a reader can sample and absorb as much from the book as their background allows. Although full appreciation of every nuance requires the reader's background to encompass the union of the authors' knowledge,

there is much to be extracted by those with rather different training. In particular, high-school geometry, basic discrete mathematics, familiarity with big-Oh notation, and a dollop of that intangible “mathematical sophistication” will make accessible at least three-quarters of the book. The remaining quarter needs, in various combinations, concepts in calculus, linear algebra, differential equations, differential geometry, and graph theory; some algorithmic techniques (e.g., linear programming and dynamic programming); and aspects of complexity theory, especially the theory of NP-completeness. In general the material is elementary, though it is sometimes challengingly intricate.

This book is not a textbook; there are no exercises, for example. It is more a “research monograph,” but we hope one with wider appeal than the usual implication of that phrase. Both of us have taught courses based on material in this book, and we know from our experiences that it can easily form the core of an undergraduate or graduate course.

Both because the field of investigation is relatively new—most of what we describe has been discovered in the last decade—and because the problems are hard, there are many unsolved, specific problems, which are known as “open problems” in the field. These problems have the unusual characteristic of being at the same time shallow and deep: shallow in the sense that they may be stated without much technical jargon (like the five mentioned above), and deep in that at least some of them apparently require significant insights to resolve. We isolate many such open problems throughout the book, boxing them to stand out and attract attention (and all are listed in the index). We should warn the reader that the status of these problems varies considerably: some are old chestnuts that have resisted repeated cracking attempts, and others occurred to us as we were writing and are quite unexplored. (The distinction is usually evident from citations to the problem originators; no citation implies we posed it.) Highlighting these problems is an invitation to work on them, not a judgment of their depth or difficulty. In fact, several were solved as the book was being written. Please check the book’s Web site¹ for the latest status.

A portion of what we describe in the book is based on published papers written with a large number of coauthors, and indeed we sometimes incorporate sections from these papers with only minor alterations. In this sense, all of our collaborators are coauthors of this book, and for the honor and thrill of working with them, we thank Pankaj K. Agarwal, Oswin Aichholzer, Rebecca Alexander, Greg Aloupis, Esther M. Arkin, Boris Aronov, Devin J. Balkcom, Nadia Benbernou, Michael A. Bender, Marshall Bern, Therese Biedl, David Bremner, Patricia Cahn, Jason H. Cantarella, Roxana Cocan, Robert Connelly, Mirela Damian, Martin L. Demaine, Satyan Devadoss, Melody Donoso, Heather Dyson, David Eppstein, Jeff Erickson, Robin Flatland, Blaise Gassend, Julie Glass, George W. Hart, Barry Hayes, Ferran Hurtado, Hayley N. Iben, Biliانا Kaneva, Eric Kuo, Stefan Langermann, Sylvain Lazard, Bin Lu, Anna Lubiw, Andrea Mantler, Henk Meijer, Joseph S. B. Mitchell, Pat Morin,

¹ <http://www.gfalop.org>; see also <http://cs.smith.edu/~orourke/TOPP> for several of the most prominent open problems. For access to the passworded area of the site use the password clamfug.

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